

RED ADAIR — TECHNICAL AFSS TEAM

AFSS Backlog Tracker

System Overview & Documentation

A complete reference for both operational users and technical staff —
covering what the system does, what it automates, and how it is built
and maintained.

Version 2.0

June 2026

Next.js 16

Google Cloud Run

SimPRO API

Live URL: <https://bryan-technical-afss-712513641417.australia-southeast1.run.app>
Contact: bryan.morales@redadair.com.au

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PART ONE

For Everyone

A plain-English explanation of what the AFSS Backlog Tracker does, what it automates, and how to use it — no technical background required.

SECTION 1 — NON-TECHNICAL

What Is This System?

The **AFSS Backlog Tracker** is an internal web dashboard built for Red Adair's fire safety compliance team. It gives management and technical staff a single, always-up-to-date view of every CFSP (Combined Fire Safety Provisions) audit job across three of the company's operating entities.

Entity	What They Do
RM AFSS (Red Men)	Primary AFSS audit team
CHUBB / AFAC	AFAC-branded audit stream
AE Evac	Evacuation procedure audits

Think of it as a live **"job board" combined with a capacity planner** — showing not just what work exists, but also how much time the team has left to do it this month.

SECTION 2 — NON-TECHNICAL

What Problem Does It Solve?

Before this system existed, answering these questions required manually logging into SimPRO (Red Adair's job management software) and pulling separate reports:

- How many audit jobs are still in the backlog?
- Which jobs are scheduled vs. unscheduled?
- How many available hours does the team have left this month?
- How much non-audit technical support work is also scheduled?
- What is the expected AFAC audit demand over the coming weeks?

The AFSS Backlog Tracker **pulls all of that data automatically** and presents it in one place, refreshed throughout the day — so management can make staffing and scheduling decisions without manual report-pulling.

SECTION 3 — NON-TECHNICAL

What Does It Automate? (Plain English)

Here is every task the system performs automatically on your behalf:

Job Data Collection

Every hour, the system quietly connects to SimPRO and downloads the latest list of pending and in-progress CFSP audit jobs for all three companies. You never have to manually export or refresh a report.

Team Availability Calculation

The system works out how many billable hours each auditor has left in the current month by starting with their standard working hours, then subtracting approved leave booked in SimPRO and NSW public holidays for the month. This gives a real, usable capacity number — not a theoretical one.

Technical Support Work Tracking

The system monitors the jobs scheduled to the tech team that are not CFSP audits (e.g., internal billable work, quality assurance tasks). It automatically categorises each job as: **Other Billable** (work billed to external customers), **Invested Time** (internal work), or **QA** (quality assurance activity).

AFAC Demand Forecasting

The system looks at what AFAC/CHUBB audit work Muhammad Soban was doing at the same time last year and uses that as a forecast for the coming weeks. This helps the team anticipate demand spikes before they arrive.

Instant Updates When Schedules Change

When someone changes a schedule in SimPRO, SimPRO sends an automatic notification to this system. The system then refreshes its data in the background — so the next time you load the dashboard, the numbers are already up to date. No manual action needed.

Login Security

The system automatically checks that anyone trying to log in is a member of the approved Red Adair Google Workspace group. People outside that group cannot access the dashboard, even if they have a Google account.

SECTION 4 — NON-TECHNICAL

Who Can Use It?

Access is restricted to members of the Google Workspace group: `technicalafss-deployment@redadair.com.au`

If you need access, ask your administrator to add your Red Adair Google account to that group.

SECTION 5 — NON-TECHNICAL

How to Log In

1. Open the live URL in your browser.
2. You will see a login page — click **Sign in with Google**.
3. Choose your `@redadair.com.au` Google account.
4. If your account is in the approved group, you will be taken straight to the dashboard.
5. If access is denied, contact your administrator.

WARMUP NOTICE

After a new deployment, the system takes approximately 10–13 minutes to warm up its data. During that time the dashboard may be slow or show a loading state. This is normal.

SECTION 6 — NON-TECHNICAL

Pages at a Glance

/ (Home)

Backlog

Full list of all pending and in-progress audit jobs, filterable by company, stage, and status. Click any row to open the job directly in SimPRO.

/dashboard

Dashboard

Management summary: job counts, team hours, technical support work, and AFAC demand forecast.

/dashboard2

Dashboard (No DATACOM)

Same as Dashboard but excludes DATACOM cost-centre jobs from the Other Billable totals.

/progress

Progress

Progress tracking view — jobs that are currently in-progress or awaiting results release.



SCREENSHOT — AFSS BACKLOG PAGE (/)

System

Insert screenshot of the main Backlog page showing the full job list table, company filter tabs, status colour coding, and column headers.

URL: <https://bryan-technical-afss-712513641417.australia-southeast1.run.app/>



SCREENSHOT — AFSS DASHBOARD (/DASHBOARD)

System

Insert screenshot of the Dashboard page showing the Supply vs Demand tables, Technical Team Supply hours, Technical Support Works, and AFAC Prospect Demand.

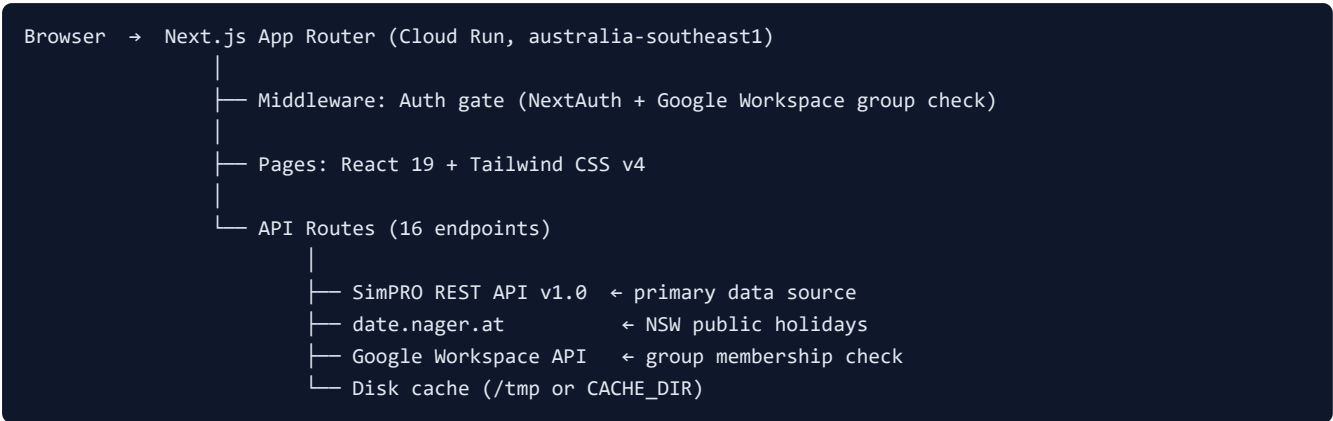
URL: <https://bryan-technical-afss-712513641417.australia-southeast1.run.app/dashboard>

PART TWO**For Technical Staff**

Architecture, API reference, caching, authentication, deployment, and all automation details for engineers maintaining this system.

SECTION 7 — TECHNICAL

Architecture Overview



Property	Value
Platform	Google Cloud Run — Node.js 22 / Alpine Linux
Container	Multi-stage Docker build (deps → builder → runner)
Next.js output mode	Standalone
Cache headers	<code>no-store</code> on all responses (caching is server-side only)
Region	australia-southeast1 (Sydney)
Timezone	AEST — UTC+10 — all dates and timestamps

SECTION 8 — TECHNICAL

Tech Stack

Layer	Technology	Version
Framework	Next.js (App Router)	16
UI Library	React	19
Styling	Tailwind CSS	v4
Language	TypeScript	5
Runtime	Node.js (Alpine Linux)	22
Authentication	NextAuth	v4
Auth Provider	Google OAuth 2.0 + Workspace Directory API	—
Primary Data Source	SimPRO REST API	v1.0
Holiday Data	date.nager.at (NSW Australia)	—
Cloud Platform	Google Cloud Run	—
Containerisation	Docker (multi-stage Alpine build)	—
PDF Generation	Puppeteer	25.x

SECTION 9 — TECHNICAL

What Gets Automated — Technical Detail

9.1 — Cache Warming on Startup (`instrumentation.ts`)

On every Cloud Run deployment, `instrumentation.ts` runs **before the server accepts any traffic** and pre-warms all caches sequentially to prevent slow first loads:

1. Fetch SimPRO job data for Company 1 (RM AFSS)
2. Wait 60 seconds — SimPRO rate limit recovery
3. Fetch SimPRO job data for Company 8 (CHUBB/AFAC)
4. Warm leave schedule, tech support, and AFAC prospect data in parallel

NOTE

Company 10 (AE Evac) is excluded from warmup due to SimPRO rate limit constraints. Total warmup time: **10–13 minutes**.



SCREENSHOT — SIMPRO JOB LIST (SOURCE FOR `/API/DATA`)

SimPRO

Insert screenshot of the SimPRO job list filtered by Stage = Pending/Progress and Technician = "A CFSP ONLY" for Company 1 (RM AFSS). This is the raw data the system reads.

SimPRO → Jobs → Filter: Stage = Pending, Technician = A CFSP ONLY, Company = RM AFSS

9.2 — SimPRO Webhook Integration (`/api/webhook/simpro`)

SimPRO sends a POST to `/api/webhook/simpro` whenever a schedule event changes. The handler:

1. Clears tech support cache files from disk
2. Kicks off a background rebuild of fresh tech support data
3. Returns 200 immediately — the rebuild runs asynchronously

9.3 — Team Availability Calculation (`/api/leave`)

- Fetches leave records for: Muhammad Soban (ID 1581), Ryan Gordon (ID 15), Josh Roger (ID 1753)
- Fetches NSW public holidays from `date.nager.at` for the current year (cached permanently on disk)
- Fallback: hardcoded NSW holidays for 2025–2026 if the API is unreachable
- Formula: remaining hours = (working days remaining – leave days – public holiday days) × daily hours
- After 3 PM, today is excluded from the remaining working days count



SCREENSHOT — SIMPRO SCHEDULE / LEAVE VIEW (SOURCE FOR `/API/LEAVE`)

SimPRO

Insert screenshot of the SimPRO schedule/calendar showing leave blocks for Muhammad Soban, Ryan Gordon, and Josh Roger. This is where leave hours are read from.

SimPRO → Schedule → Filter by Staff: Soban / Gordon / Roger — look for Annual Leave / Sick Leave blocks

9.4 — Technical Support Work Categorisation (`/api/tech-support`)

Mirrors SimPRO's "Schedule Breakdown" report exactly. For each scheduled block in the tech team's calendar:

- Filters by 38 defined AFSS cost centre IDs
- Classifies customer type: Internal (Redmen Fire, AFAC, Adair, Z SAFE) → **Invested Time**; External → **Other Billable**; QA cost centres → **Quality Assurance**
- Cache TTLs: OB/IT = 5 minutes, QA = 1 hour
- Force rebuild: append `?force=1`
- Debug export: `/api/tech-support?debug=list&year=Y&month=M`



SCREENSHOT — SIMPRO SCHEDULE BREAKDOWN REPORT (SOURCE FOR `/API/TECH-SUPPORT`)

SimPRO

AFSS Backlog Tracker — System Overview — Red Adair — Confidential — June 2026

Insert screenshot of the SimPRO Schedule Breakdown report filtered by the 3 tech team members for the current month. This is the exact report the system replicates automatically.

SimPRO → Reports → Schedule Breakdown → Staff: Soban, Gordon, Roger → Date: current month

9.5 — AFAC Prospect Demand Forecasting (`/api/afac-prospect`)

- Uses Muhammad Soban's schedule from the same date range last year as a proxy for expected demand
- Supports date exclusions stored in `data/afac-exclusions.json`
- Cache TTL: 30 minutes
- Returns: job count, estimated hours, date range



SCREENSHOT — SIMPRO SOBAN SCHEDULE LAST YEAR (SOURCE FOR `/API/AFAC-PROSPECT`)

SimPRO

Insert screenshot of Muhammad Soban's schedule in SimPRO for the same period last year. This is the historical data the system uses to forecast AFAC demand.

SimPRO → Schedule → Staff: Muhammad Soban → Date: same month, previous year → filter AFSS-Audit jobs only

9.6 — SimPRO Rate Limit Handling

SimPRO enforces strict per-token rate limits (HTTP 429). The app implements:

- Exponential backoff: 1s → 2s → 4s → 8s → ... up to 8 retry attempts
- Sequential company fetches (not parallel) during warmup to avoid exhausting the retry budget

9.7 — Authentication Middleware (`middleware.ts`)

All routes except `/login` and `/api/auth/*` are gated. The middleware:

1. Checks for a valid NextAuth session cookie
2. Validates the user's email is a member of `technicalafss-deployment@redadair.com.au` via the Google Workspace Directory API
3. Uses service account `afss-group-checker@technical-afss.iam.gserviceaccount.com` with subject impersonation for Directory API calls

SECTION 10 — TECHNICAL

API Routes Reference

Endpoint	Method	Description	Cache TTL
<code>/api/data?company=N&stage=S</code>	GET	AFSS audit job list from SimPRO	1 hour
<code>/api/data?force=1</code>	GET	Force live rebuild of job cache	—
<code>/api/leave</code>	GET	Team leave + NSW public holidays	1 hour
<code>/api/tech-support</code>	GET	OB / IT / QA stats	5 min (OB/IT), 1 hr (QA)
<code>/api/tech-support?force=1</code>	GET	Force rebuild of tech support cache	—
<code>/api/tech-support?debug=list&year=Y&month=M</code>	GET	Full schedule block list (no cache)	—
<code>/api/afac-prospect</code>	GET	AFAC prospect demand forecast	30 min

Endpoint	Method	Description	Cache TTL
<code>/api/afac-exclusions</code>	GET / POST	Read or update AFAC excluded dates	No cache
<code>/api/intercompany</code>	GET / POST	Read or update intercompany hours	No cache
<code>/api/extra-team</code>	GET	Team member metadata	—
<code>/api/extra-team/search</code>	GET	Search team members	—
<code>/api/jobs</code>	GET	Job detail	—
<code>/api/progress</code>	GET	Progress tracking data	—
<code>/api/warmup</code>	GET	Manually trigger full cache warm	—
<code>/api/webhook/simpro</code>	POST / GET	SimPRO schedule change webhook	—
<code>/api/auth/[...nextauth]</code>	GET / POST	NextAuth OAuth callbacks	—

Company values: `1` = RM AFSS | `8` = CHUBB/AFAC | `10` = AE Evac

Stage values: `Pending` or `Progress`

SECTION 11 — TECHNICAL

Caching Strategy

All caching is disk-based. In development the location defaults to `os.tmpdir()`. In production, `CACHE_DIR` points to a GCS-mounted volume for persistence across container restarts.

Cache	File Pattern	TTL
AFSS audit jobs	<code>/tmp/afss-v4-{company}-{stage}-cache.json</code>	1 hour
Team leave + holidays	<code>/tmp/afss-leave-cache-v2.json</code>	1 hour
Tech support OB/IT	<code>/tmp/afss-tech-support-v84-{year}-{month}.json</code>	5 minutes
Tech support QA	<code>/tmp/afss-qa-v1.json</code>	1 hour
AFAC prospect demand	<code>/tmp/afss-afac-prospect-{year}-{month}.json</code>	30 minutes
NSW public holidays	<code>/tmp/afss-public-holidays-nsw-{year}.json</code>	Permanent (never evicted)

GCS CACHE STATUS

`gcsCache.ts` is present in the codebase but currently disabled. The disk cache is the active caching layer.

SECTION 12 — TECHNICAL

Authentication & Access Control

Step	Mechanism
Login	Google OAuth 2.0 via NextAuth v4
Group check	Google Workspace Directory API (<code>admin.googleapis.com</code>)
Allowed group	<code>technicalafss-deployment@redadair.com.au</code>
Service account	<code>afss-group-checker@technical-afss.iam.gserviceaccount.com</code>

Step	Mechanism
Admin impersonation	<code>bryan.morales@redadair.com.au</code>
Session storage	NextAuth JWT (cookie-based)
Middleware scope	All routes except <code>/login</code> and <code>/api/auth/*</code>

SECTION 13 — TECHNICAL

Environment Variables

```
# SimPRO
SIMPRO_BASE_URL=https://redmen.simprosuite.com
SIMPRO_TOKEN=<bearer token>

# Google OAuth (NextAuth)
GOOGLE_CLIENT_ID=...
GOOGLE_CLIENT_SECRET=...

# NextAuth
NEXTAUTH_SECRET=...
NEXTAUTH_URL=http://localhost:3000
NEXTAUTH_URL_PRODUCTION=https://<cloud-run-url>

# Google Workspace service account
GOOGLE_SERVICE_ACCOUNT_EMAIL=afss-group-checker@technical-afss.iam.gserviceaccount.com
GOOGLE_SERVICE_ACCOUNT_PRIVATE_KEY=-----BEGIN PRIVATE KEY-----...
GOOGLE_ADMIN_EMAIL=bryan.morales@redadair.com.au

# Optional – production only
CACHE_DIR=/path/to/gcs-volume
```

SECURITY NOTE

All secrets are set in Cloud Run as secret environment variables. Never commit them to source control. The SIMPRO_TOKEN is automatically stripped of any leading UTF-8 BOM character on load.

SECTION 14 — TECHNICAL

Deployment Process

Pre-Deploy — Sync Persistent Data from Production

```
Invoke-WebRequest -Uri "https://<prod-url>/api/intercompany" -OutFile "data\intercompany.json"
Invoke-WebRequest -Uri "https://<prod-url>/api/afac-exclusions" -OutFile "data\afac-exclusions.json"
git add data/
git commit -m "Sync persistent data before deploy"
```

Deploy to Google Cloud Run

```
gcloud run deploy bryan-technical-afss \
  --source . \
  --region australia-southeast1 \
  --timeout=300 \
  --min-instances=1 \
  --quiet
```

Post-Deploy Checklist

- 1. Wait **10–13 minutes** for `instrumentation.ts` warmup to complete
- 2. Reload the dashboard and verify data is displaying correctly
- 3. Check Cloud Run logs if the warmup appears to stall

Local Development

```
npm install
cp .env.local.example .env.local # fill in credentials
npm run dev # starts on http://localhost:3000
```

SECTION 15 — TECHNICAL

Data Files & Key Business Rules

Persistent Data Files (`data/`)

These JSON files are tracked in git and hold state that must survive between deployments. Because Cloud Run containers are ephemeral, any data written to `/tmp` at runtime is lost on restart — these three files are the only persistent state, and must be synced before each deployment (see Section 14).

File	Purpose	Editable Via
<code>data/intercompany.json</code>	Intercompany hours for RM, AE, and FIA	Dashboard UI or <code>POST /api/intercompany</code>
<code>data/afac-exclusions.json</code>	Dates excluded from AFAC demand forecasts	Dashboard UI or <code>POST /api/afac-exclusions</code>
<code>data/extra-team-members.json</code>	Supplementary team metadata	Direct file edit, then redeploy

Key Business Rules — Quick Reference

Rule	Detail
Job filter — Companies 1 & 10	Only jobs assigned to technician "A CFSP ONLY" (ID 1126)
Job filter — Company 8	All jobs, no technician filter
Default job price	\$330 if SimPRO total is blank
Default job hours	2 hours if SimPRO estimated hours = 0
Invoice display	Ex-tax amounts
Status — Progress (Co 1 & 8)	Attendance complete (grey)
Status — Progress (Co 10 AE Evac)	Audit booked but not yet complete
Status — Scheduled	Job has a scheduled date set (green)
Status — Awaiting Client Info	Blue
Status — Tentative	Yellow
Timezone	All dates/times in AEST (UTC+10)
Cost centre count	38 AFSS cost centres mapped by ID
Internal clients (Invested Time)	Redmen Fire, Z SAFE OS, Adair Operation, AFAC (case-insensitive match)
Team member IDs	Soban: 1581 Ryan: 15 Josh: 1753 CFSP tech: 1126